

AcePharm — Mobile App Brief

Cross-platform companion mobile application for iOS and Android (Flutter / Dart)

Prepared for	Muhammad Numan Latif / Mobile Engineering Team
Client & Product	Takween Centre UK Ltd · AcePharm Mobile App (iOS & Android)
Project type	Native Mobile App Delivery — High-Fidelity Flutter Companion Application
Reference documents	AcePharm Developer Brief v2.1, Mobile Brief v1.0, Interactive HTML Prototype
Core Stack	Flutter 3.x / Dart (iOS & Android) · Shared Next.js Backend & Prisma API · Auth.js API Authentication · Hive / SQLite Offline Storage · Firebase Crashlytics & Analytics
Core feature set	Mobile-Optimized Practice Engine (Learn & Timed) · Native Confidence Capture · Full Clinical Explanations · Contextual Ask Ace RAG Client · 5-Metric Offline-Resilient Progress Tracking · Subscription Status & Apple/Google IAP Alignment
Distribution (QA)	Google Play Console Internal Testing (APK) & Apple TestFlight ahead of production store release

Purpose

The AcePharm Mobile App is the official cross-platform companion application for UK pharmacy undergraduates and pre-registration trainees, built to bring high-yield revision and clinical practice directly to iOS and Android devices. Operating against the unified AcePharm backend, the mobile application provides frictionless question practice on the go, allowing students to convert spare commute and hospital placement moments into high-retention learning.

This brief sets the formal production standards for the mobile build. The mobile application is not a stripped-back or simplified utility; it implements the complete clinical question engine, the rigorous two-tier attempt tracking, and the grounded AI tutor layer with native mobile ergonomics. Crucially, the app must demonstrate flawless resilience against intermittent hospital ward connectivity and zero login disruption.

Design & UI — Match the Existing Spec

Mobile visual identity must strictly map to the AcePharm Design System tokens and interactive prototype specifications. Developers must translate these values directly into the Flutter ThemeData:

- **Brand Color Tokens:** Primary Ace Indigo (#4F46E5), Deep Indigo (#3730A3), Ink primary text (#111827), Slate secondary text (#64748B), Canvas background (#F8FAFC), Surface card container (FFFFFF), Border stroke (#E2E8F0). Clinical Teal (#0F766E) for references, and unambiguous semantic alert colors.
- **Typography:** Geist Sans (or system sans-serif fallback) for general interface copy and clinical rationales; Geist Mono for all pharmaceutical calculations, formulas, timers, and question public IDs.
- **Ergonomics:** 8-point layout grid, 12–16px card border radii, subtle 1px border strokes rather than heavy shadows, and strict adherence to a 44×44pt minimum touch target for all interactive elements to support rapid one-handed mobile revision.

What to Build

1. Native Authentication & 5-Step Mobile Onboarding

- Synchronized session authentication utilizing the web platform's Auth.js token endpoints (email/password and magic-link verification). Secure token caching via flutter_secure_storage.
- Fluid, touch-optimized 5-step onboarding sequence capturing stage, target exam date, daily question goal, and UK pharmacy school affiliation, fully synchronized with the central web database.

2. Mobile Dashboard & Smart Recommendation Engine

- Home dashboard surfacing the learner's dynamic 'Recommended Next Session' with transparent reasoning (e.g. 'Strengthen low-confidence antimicrobial questions').
- Daily goal progress rings, meaningful revision streaks, weekly Ace clinical insights, and active subscription status banners.

3. Question Engine & Clinical Practice Loop (Core Feature)

- **Session Configuration:** Intuitive mobile bottom-sheet selectors for clinical categories, high-risk drug flags, question count, and mode (Learn vs. Timed exam conditions).

- **Strict Clinical Rules:** Mandatory confidence rating (Low, Medium, High) captured before answer submission. Zero color feedback on options prior to tapping Submit. Post-submission feedback must display distinct icons and textual status alongside colour. Auto-advance is strictly disabled.
- **Explanation Layout:** Collapsible per-option rationales, comprehensive subtopic clinical revision notes, personal sticky note taking, bookmarking, and instant question error reporting attributed to 'AcePharm Clinical Editorial Team'.

4. Progress Tracking & Topic Map Progression

- Clean visual presentation of the five uncompromised performance metrics: First-Attempt Accuracy, Practice Accuracy, Repeat Accuracy, Confidence Calibration, and Curriculum Coverage.
- Interactive mobile Topic Map enforcing the strict six-tier status hierarchy: *Not Started* → *First Pass* → *Needs Attention* → *Developing* → *Secure* → *Due for Review*.

5. Contextual AI Tutor ('Ask Ace') Client

- Contextual bottom-sheet or inline chat panel beneath question explanations allowing learners to query specific drug mechanisms or calculations.
- Streaming RAG responses strictly grounded in AcePharm reviewed content, displaying explicit citations. Ace must decline out-of-domain queries and patient-specific inquiries plainly and gracefully.

6. Account, Subscription Display & In-App Purchase Architecture

- Account profile management, password resets, and subscription tier overview (Plan name, renewal date, and billing price).
- Strategic implementation of subscription access: displaying web-purchased entitlement status, with clear architectural isolation should native Apple In-App Purchase (StoreKit) / Google Play Billing be activated in Phase 2.

Handling No Login and No Internet (Offline Resilience)

Pharmacy students frequently revise inside hospital basements, clinical wards, and transit areas with intermittent or zero internet connectivity. The application must treat offline states as a standard operational reality:

- **Crash-Proof Auth Guard:** Expired tokens or network dropouts must never cause application crashes or white-screen deadlocks. Present an informative offline bar with automatic background reconnection.
- **Local Attempt Queueing:** If an answer is submitted while offline, the attempt payload (selected option, confidence score, time elapsed, timestamp) must be cached locally in SQLite/Hive and automatically synced to the server once connection resumes with zero data loss.
- **Cached Review Assets:** Active revision sessions, subtopic notes, and dashboard analytics must cache their latest state locally so revision can continue uninterrupted without connectivity.

Technical Requirements

- **Framework & Tooling:** Flutter 3.x, Dart 3.x, targeting modern iOS (iOS 16+) and Android (API level 26+). Strict linting via standard analysis options.
- **API Synchronization:** Native integration with the central Next.js REST API. No duplicate or competing backend infrastructure. Full compatibility with Auth.js session cookies/JWT tokens.
- **Security & Environment Isolation:** Zero hardcoded API secrets or backend keys. Configurable environment configurations (staging vs. production flavors). Token storage in iOS Keychain / Android Keystore.
- **Distribution Standard:** Automated release packaging via Fastlane/GitHub Actions, generating signed APK/AAB builds for Google Play Console Internal Testing and TestFlight builds for iOS reviewers.

What to Submit

When submitting mobile releases for review, complete the formal Mobile Submission Document containing each of the following twelve verification fields:

#	Field	What to Include
1	App Access & Distribution	Play Console Internal Testing invite link and downloadable signed APK; Apple TestFlight public invite link or detailed staging test status.
2	Feature & Scope Mapping	Clear breakdown of web spec features translated to mobile, specifically noting any deferred Ace Hub modules.
3	Subscription & IAP Approach	Technical implementation details of account entitlement validation versus native in-app billing.

4	GitHub Repository	Full repository link, test coverage stats, and incremental commit history reflecting clean modular architecture.
5	Loom Screen Demonstration	4–6 minute video demonstrating: cold app start, onboarding, building a session, answering with confidence rating, explanation display, Ask Ace query, progress screen, and offline flight-mode attempt sync.
6	What Was Built	5 to 10 line plain-English overview of user-facing features and mobile capabilities delivered.
7	Tools & Libraries Used	Inventory of core Flutter packages (e.g. Riverpod/Bloc, Dio, Hive, flutter_secure_storage) with technical justification.
8	Problems & Resolutions	Specific engineering challenges encountered during mobile development and exact architectural solutions implemented.
9	How to Test	Step-by-step test instructions, including seed login credentials and specific test cases for offline queuing.
10	Known Limitations	Current edge cases or platform-specific polish items scheduled for subsequent build sprints.
11	Path to Production Plan	Full written response to the architectural roadmap below, structured in clear paragraphs.
12	Submission Date	Formal calendar date of submission.

QA Review Criteria

#	Assessment Point
1	Is confidence captured strictly before submission, with zero option color indication before submit and multi-modal feedback afterwards?
2	Does every option display an independent clinical rationale attributed to the AcePharm Clinical Editorial Team?
3	Are the five progress metrics clearly distinct, with First-Attempt accuracy most prominent and topic statuses following the exact 6-stage order?
4	Does the contextual Ask Ace feature respond strictly from retrieved clinical content with citations and decline out-of-domain prompts?
5	Does the app operate gracefully under offline flight mode, caching submitted answers and syncing without data loss upon reconnection?
6	Does the mobile interface accurately implement the AcePharm design system tokens (Indigo #4F46E5, 44pt touch targets, Geist typography)?
7	Is the codebase well-structured with clear separation of UI, business logic, and networking layers?
8	Can QA reviewers install and launch the application seamlessly via shareable APK or TestFlight without personal developer credentials?

Path to Production

Moving the AcePharm mobile application into active store distribution requires disciplined execution across shared architecture, API synchronization, and app store compliance. As the third mobile product built under the Takween Digital umbrella (alongside Prayer Times and Takween Tutors), AcePharm should leverage a unified internal Flutter foundation: shared networking interceptors, unified design system primitives, and standard offline sync handlers to prevent redundant engineering.

Developing mobile features ahead of web milestones requires a rigorous contract-first methodology. The mobile team must build against versioned OpenAPI specifications and structured mock fixtures matching the web platform's data models. This ensures zero architectural rework once Milestone 3 (the live question engine API) and Milestone 4 (progress analytics) deploy to staging.

Finally, navigating Apple App Store Review Guideline 3.1.1 and Google Play Billing requirements mandates a clear monetization roadmap. The v1 release displays account entitlements for subscriptions purchased via the web platform under the 'Reader' app exception. If direct mobile purchasing is introduced in future releases, the team must implement native StoreKit and Google Play Billing alongside receipt validation webhooks, account restoration buttons, and localized terms of service.